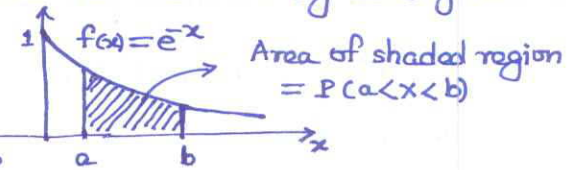


• A random variable X is said to be continuous if there exists a nonnegative function $f(x)$ such that the cumulative distribution function $F(x) = \int_{-\infty}^x f(t) dt$ for every real number x . The function $f(x)$ is called the probability density function of X . The probability density function (PDF) satisfies (i) $f(x) \geq 0$ for all x .

(ii) $\int_{-\infty}^{\infty} f(t) dt = 1$
 (iii) $\int_a^b f(t) dt = P(a \leq X \leq b)$

• Ex. Suppose the time to failure of the cathod ray tube has the following density function:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



This PDF is called the exponential density (with mean 1), and the experimental evidence has indicated that it is appropriate to describe the time to failure (a real-world occurrence) for some types of components. Suppose we want to find $P(X \geq 100 \text{ hours})$:

$$P(X \geq 100) = \int_{100}^{\infty} \lambda e^{-\lambda t} dt = e^{-100\lambda}$$

• Determine $P(X \geq 100 | X > 99)$, the probability the tube lives at least 100 hours given that it has lived beyond 99 hours.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\int_{100}^{\infty} \lambda e^{-\lambda t} dt}{\int_{99}^{\infty} \lambda e^{-\lambda t} dt} = \frac{e^{-100\lambda}}{e^{-99\lambda}} = e^{-\lambda} = P(X > 1)$$

$$P(X > t+h | X > h) = P(X > t) \text{ for any } t > 0, h > 0$$

• Expectation: If X is a random variable and $Y = g(X)$ is a function of X , then the expected of $g(X)$ is defined as follows:

$$E[g(X)] = \begin{cases} \sum_i g(x_i) p(x_i) & \text{for } X \text{ discrete,} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & \text{for } X \text{ continuous.} \end{cases}$$

If $g(x) = x$, $E(X) = \begin{cases} \sum_i x_i p(x_i) & \text{for } X \text{ discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{for } X \text{ continuous} \end{cases}$

The expected value of the random variable X is called the mean and it is denoted by μ .

If $g(X) = (X - \mu)^2$ then

$$E[g(X)] = E(X - \mu)^2 = \sigma^2 \leftarrow \text{the variance of the RV } X. \\ = E(X^2) - \{E(X)\}^2$$

Properties: $E(aX + b) = aE(X) + b$, $a > 0$
 $V(aX + b) = a^2 V(X)$

• The Normal Distribution has been used as a model for diverse phenomena such as a person's height, the distribution of IQ scores, and the velocity of a gas molecule.